S.No: 3	Exp. Name: ***Minimum Candy Distribution***	Date:2025-06-24

Aim:

There are N children standing in a line, each with a given rating value. You need to distribute candies to the children while following these rules:

- Each child must get at least one candy.
- A child with a higher rating than their neighbor(s) must receive more candies than them.

Write a C program to determine the minimum number of candies required to satisfy these conditions.

Input Format:

- The first line contains a positive integer N (number of children).
- The second line contains N space-separated integers, representing the rating values of each child.

Output Format:

- Print a single integer representing the minimum number of candies required.

Source Code:

minimum_candies.c

```
//  Write your code here
#include<stdio.h>
int max(int a, int b){
   if (a > b) {
      return a;
   }
   else
      return b;
}
int main() {
   int n;
   int sum = 0;
   scanf("%d",&n);
   int rating[n];
   for (int i = 0; i < n; i++) {
      scanf("%d",&rating[i]);
   }
   int candy[n];
   for(int i = 0; i < n; i++){
      candy[i] = 1;
   }

   for (int i = 1; i < n; i++) {
      if(rating[i] > rating[i-1]) {
         candy[i] = candy[i-1] + 1;
      }
   }
   for(int i = n-2; i >= 0; i--){
      if(rating[i]>rating[i+1])
         candy[i] = max(candy[i], candy[i+1]+1);
   }
   for(int i = 0; i < n; i++){
      sum = sum + candy[i];
```

```
    }

    printf("%d\n",sum);
    return 0;


}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
5
1 2 3 4 5
15

Test Case - 2
User Output
4
4 3 2 1
10

Test Case - 3
User Output
5
2 2 2 2 2
5